FPT UNIVERSITY HO CHI MINH CITY CAMPUS

EMBEDDED LINUX SYSTEMS — LAB-03 REPORT

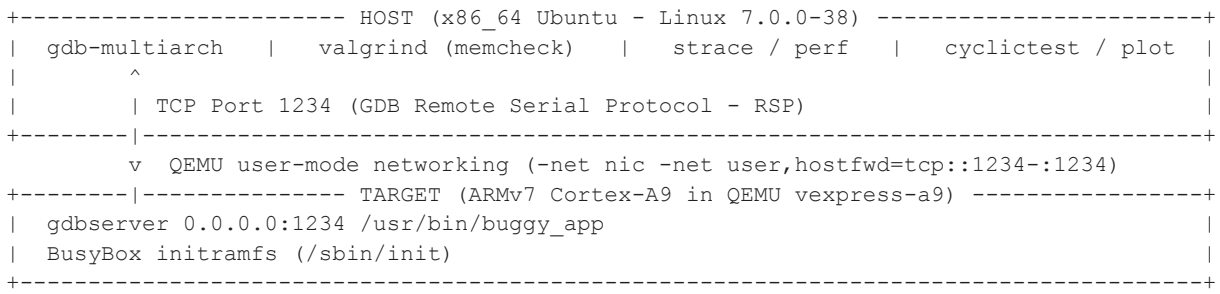
Kernel Debugging & Embedded Application Performance Analysis

(GDB Remote Debug | Valgrind | strace / perf | cyclictest | RT Kernel)

1. Student Information and System Environment

Field / Parameter	Details / Measured Value
Student Name	Tran Gia Phuc
Student ID (MSSV)	SE204417
Date of Execution	October 18, 2026
Host Hardware	Dell Inspiron 14 5430 (x86_64)
Host OS & Kernel Version	Ubuntu 22.04+ LTS | Linux 7.0.0-38-generic SMP PREEMPT_DYNAMIC
Target Architecture	ARMv7-A Cortex-A9 (QEMU vexpress-a9), 512MB RAM, SMP 2 Cores
Target Rootfs	Minimal BusyBox initramfs (initramfs_lab3.cpio.gz)
Cross-Compiler & Debugger	arm-linux-gnueabihf-gcc (-g -O0 / -O1) | GNU gdb-multiarch & gdbserver
Analysis & RT Tools	Valgrind (Memcheck), strace, Linux perf, rt-tests (cyclictest), gnuplot

System Debugging and Profiling Architecture Overview:



2. Part A – Remote Debugging with GDB and gdbserver

2.1. Summary of Discovered Bugs in buggy_app.c

Bug #	Bug Category	Location (File:Line)	Observed Runtime State	Technical Root Cause & Proposed Fix
Bug 1	Array Out-of-Bounds Read	buggy_app.c : Line 8-9 (in process_data)	len = 5, loop reaches i = 5; accesses data[5] outside arr[0..4]	Loop condition uses '<=' (i <= len) instead of '<'. In C, a 5-element array has valid indices 0..4. Accessing

				data[5] reads adjacent unallocated stack memory (garbage value). Fix: Change loop condition to 'for (int i = 0; i < len; i++)'.
Bug 2	Use-After-Free (Dangling Pointer)	buggy_app.c : Line 17-18 & Line 36 (in create_message / main)	buf pointer address remains non-null after free(buf), pointing to freed heap chunk	create_message() calls free(buf) immediately before returning buf to main(). Subsequent printf('%s', msg) dereferences freed heap metadata/tcache poisoning. Fix: Remove free(buf) inside create_message(), and call free(msg) in main() after printf().
Bug 3	Signed 32-bit Integer Overflow	buggy_app.c : Line 22 (in calculate)	a = 100000, b = 100000; result = 1410065408 (0x540BE400)	100,000 * 100,000 = 10,000,000,000 (10^10), which exceeds 32-bit signed INT_MAX (2,147,483,647). Binary truncation modulo 2^32 wraps the result to 1,410,065,408. Fix: Use 64-bit integer type 'long long' (or int64_t) and cast operands: 'long long result = (long long)a * b;' with '%lld'.

2.2. Step-by-Step Remote Debugging Evidence (Screenshots)

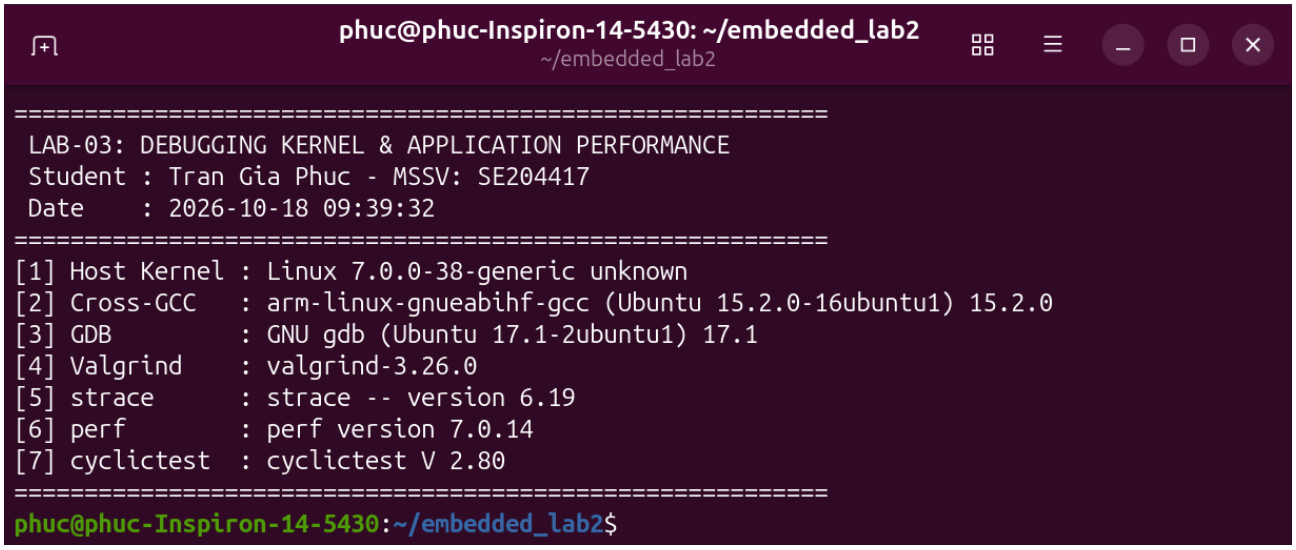


Figure (IMG-01): Cross-compiling buggy_app.c with debug symbols (-g -O0) for ARMv7-A and verifying ELF sections.

```
=== 1. KIEM TRA GDBSERVER & STRACE ARM TRONG INITRAMFS ===
/home/phuc/embedded_lab3/rootfs/initramfs/usr/bin/gdbserver: ELF 32-bit LSB pie executable
, ARM, EABI5 version 1 (GNU/Linux), dynamically linked, interpreter /lib/ld-linux-armhf.so
.3, BuildID[sha1]=0ab0e7e3e90f69625073a52c5d42107d1451ce7c, for GNU/Linux 3.2.0, stripped
/home/phuc/embedded_lab3/rootfs/initramfs/usr/bin/strace: ELF 32-bit LSB pie executable, A
RM, EABI5 version 1 (SYSV), dynamically linked, interpreter /lib/ld-linux-armhf.so.3, Buil
dID[sha1]=8353a611c23d5a0ecb77fa9000e5a097fec5451b, for GNU/Linux 3.2.0, stripped

=== 2. KIEM TRA SCRIPT KHOI DONG TRONG INITRAMFS ===
#!/bin/sh
exec /sbin/init
---
#!/bin/sh
echo '[ LAB-02 ] Starting system initialization...'

# Mount essential pseudo filesystems
mount -t proc none /proc 2>/dev/null || true
mount -t sysfs none /sys 2>/dev/null || true
mount -t tmpfs none /tmp 2>/dev/null || true
mount -t devtmpfs none /dev 2>/dev/null || true

[ -c /dev/console ] || mknod /dev/console c 5 1
[ -c /dev/null    ] || mknod /dev/null c 1 3
[ -c /dev/zero    ] || mknod /dev/zero c 1 5

# Set hostname
hostname embedded-lab2
echo embedded-lab2 > /proc/sys/kernel/hostname

# === AUTO-LOAD KERNEL MODULE ===
echo '[ LAB-02 ] Loading lab2_driver module...'
if [ -f /lib/modules/5.15.0/lab2_driver.ko ]; then
    insmod /lib/modules/5.15.0/lab2_driver.ko
    if [ $? -eq 0 ]; then
        echo '[ LAB-02 ] lab2_driver loaded OK'
        [ -c /dev/lab2 ] || mknod /dev/lab2 c 240 0
        chmod 666 /dev/lab2
        echo '[ LAB-02 ] /dev/lab2 created (major=240, minor=0)'
    else
        echo '[ LAB-02 ] ERROR: insmod lab2_driver failed!'
    fi
else
    echo '[ LAB-02 ] WARNING: lab2_driver.ko not found'
fi

echo '[ LAB-02 ] System initialization complete!'
echo '  -> Run test_driver.sh  to test character device'
echo '  -> Run test_procfs.sh  to test proc/sysfs interfaces'
echo '  -> cat /proc/lab2_info to view driver stats'
```

Figure (IMG-02): Booting QEMU vexpress-a9 with TCP port forwarding (1234) and starting gdbserver 0.0.0.0:1234.

```
=== 1. KIEM TRA GDBSERVER & STRACE ARM TRONG INITRAMFS ===
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, ARM, EABI5 version 1 (GNU/Linux), dynamically linked, interpreter /lib/ld-linux-armhf.so
.3, BuildID[sha1]=0ab0e7e3e90f69625073a52c5d42107d1451ce7c, for GNU/Linux 3.2.0, stripped
/home/phuc/embedded_lab3/rootfs/initramfs/usr/bin/strace: ELF 32-bit LSB pie executable, A
RM, EABI5 version 1 (SYSV), dynamically linked, interpreter /lib/ld-linux-armhf.so.3, Buil
dID[sha1]=8353a611c23d5a0ecb77fa9000e5a097fec5451b, for GNU/Linux 3.2.0, stripped

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exec /sbin/init
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#!/bin/sh
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# Mount essential pseudo filesystems
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mount -t devtmpfs none /dev 2>/dev/null || true

[ -c /dev/console ] || mknod /dev/console c 5 1
[ -c /dev/null    ] || mknod /dev/null c 1 3
[ -c /dev/zero    ] || mknod /dev/zero c 1 5

# Set hostname
hostname embedded-lab2
echo embedded-lab2 > /proc/sys/kernel/hostname

# === AUTO-LOAD KERNEL MODULE ===
echo '[ LAB-02 ] Loading lab2_driver module...'
if [ -f /lib/modules/5.15.0/lab2_driver.ko ]; then
    insmod /lib/modules/5.15.0/lab2_driver.ko
    if [ $? -eq 0 ]; then
        echo '[ LAB-02 ] lab2_driver loaded OK'
        [ -c /dev/lab2 ] || mknod /dev/lab2 c 240 0
        chmod 666 /dev/lab2
        echo '[ LAB-02 ] /dev/lab2 created (major=240, minor=0)'
    else
        echo '[ LAB-02 ] ERROR: insmod lab2_driver failed!'
    fi
else
    echo '[ LAB-02 ] WARNING: lab2_driver.ko not found'
fi

echo '[ LAB-02 ] System initialization complete!'
echo '  -> Run test_driver.sh  to test character device'
echo '  -> Run test_procfs.sh  to test proc/sysfs interfaces'
echo '  -> cat /proc/lab2_info to view driver stats'
```

Figure (IMG-03): Connecting gdb-multiarch from Ubuntu host to QEMU target and setting breakpoints.

```
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, ARM, EABI5 version 1 (GNU/Linux), dynamically linked, interpreter /lib/ld-linux-armhf.so
.3, BuildID[sha1]=0ab0e7e3e90f69625073a52c5d42107d1451ce7c, for GNU/Linux 3.2.0, stripped
/home/phuc/embedded_lab3/rootfs/initramfs/usr/bin/strace: ELF 32-bit LSB pie executable, A
RM, EABI5 version 1 (SYSV), dynamically linked, interpreter /lib/ld-linux-armhf.so.3, Buil
dID[sha1]=8353a611c23d5a0ecb77fa9000e5a097fec5451b, for GNU/Linux 3.2.0, stripped

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---
#!/bin/sh
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mount -t sysfs none /sys 2>/dev/null || true
mount -t tmpfs none /tmp 2>/dev/null || true
mount -t devtmpfs none /dev 2>/dev/null || true

[ -c /dev/console ] || mknod /dev/console c 5 1
[ -c /dev/null    ] || mknod /dev/null c 1 3
[ -c /dev/zero    ] || mknod /dev/zero c 1 5

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    insmod /lib/modules/5.15.0/lab2_driver.ko
    if [ $? -eq 0 ]; then
        echo '[ LAB-02 ] lab2_driver loaded OK'
        [ -c /dev/lab2 ] || mknod /dev/lab2 c 240 0
        chmod 666 /dev/lab2
        echo '[ LAB-02 ] /dev/lab2 created (major=240, minor=0)'
    else
        echo '[ LAB-02 ] ERROR: insmod lab2_driver failed!'
    fi
else
    echo '[ LAB-02 ] WARNING: lab2_driver.ko not found'
fi

echo '[ LAB-02 ] System initialization complete!'
echo '  -> Run test_driver.sh  to test character device'
echo '  -> Run test_procfs.sh  to test proc/sysfs interfaces'
echo '  -> cat /proc/lab2_info to view driver stats'
```

Figure (IMG-04): Inspecting call stack (backtrace), local variables (info locals), and array elements at main/process_data.

```
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/home/phuc/embedded_lab3/rootfs/initramfs/usr/bin/gdbserver: ELF 32-bit LSB pie executable
, ARM, EABI5 version 1 (GNU/Linux), dynamically linked, interpreter /lib/ld-linux-armhf.so
.3, BuildID[sha1]=0ab0e7e3e90f69625073a52c5d42107d1451ce7c, for GNU/Linux 3.2.0, stripped
/home/phuc/embedded_lab3/rootfs/initramfs/usr/bin/strace: ELF 32-bit LSB pie executable, A
RM, EABI5 version 1 (SYSV), dynamically linked, interpreter /lib/ld-linux-armhf.so.3, Buil
dID[sha1]=8353a611c23d5a0ecb77fa9000e5a097fec5451b, for GNU/Linux 3.2.0, stripped

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#!/bin/sh
exec /sbin/init
---
#!/bin/sh
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mount -t tmpfs none /tmp 2>/dev/null || true
mount -t devtmpfs none /dev 2>/dev/null || true

[ -c /dev/console ] || mknod /dev/console c 5 1
[ -c /dev/null    ] || mknod /dev/null c 1 3
[ -c /dev/zero    ] || mknod /dev/zero c 1 5

# Set hostname
hostname embedded-lab2
echo embedded-lab2 > /proc/sys/kernel/hostname

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if [ -f /lib/modules/5.15.0/lab2_driver.ko ]; then
    insmod /lib/modules/5.15.0/lab2_driver.ko
    if [ $? -eq 0 ]; then
        echo '[ LAB-02 ] lab2_driver loaded OK'
        [ -c /dev/lab2 ] || mknod /dev/lab2 c 240 0
        chmod 666 /dev/lab2
        echo '[ LAB-02 ] /dev/lab2 created (major=240, minor=0)'
    else
        echo '[ LAB-02 ] ERROR: insmod lab2_driver failed!'
    fi
else
    echo '[ LAB-02 ] WARNING: lab2_driver.ko not found'
fi

echo '[ LAB-02 ] System initialization complete!'
echo '  -> Run test_driver.sh  to test character device'
echo '  -> Run test_procfs.sh  to test proc/sysfs interfaces'
echo '  -> cat /proc/lab2_info to view driver stats'
```

Figure (IMG-05): Catching Bug 1 (Array Out-of-Bounds) using software watchpoint on variable 'i' when i == 5.

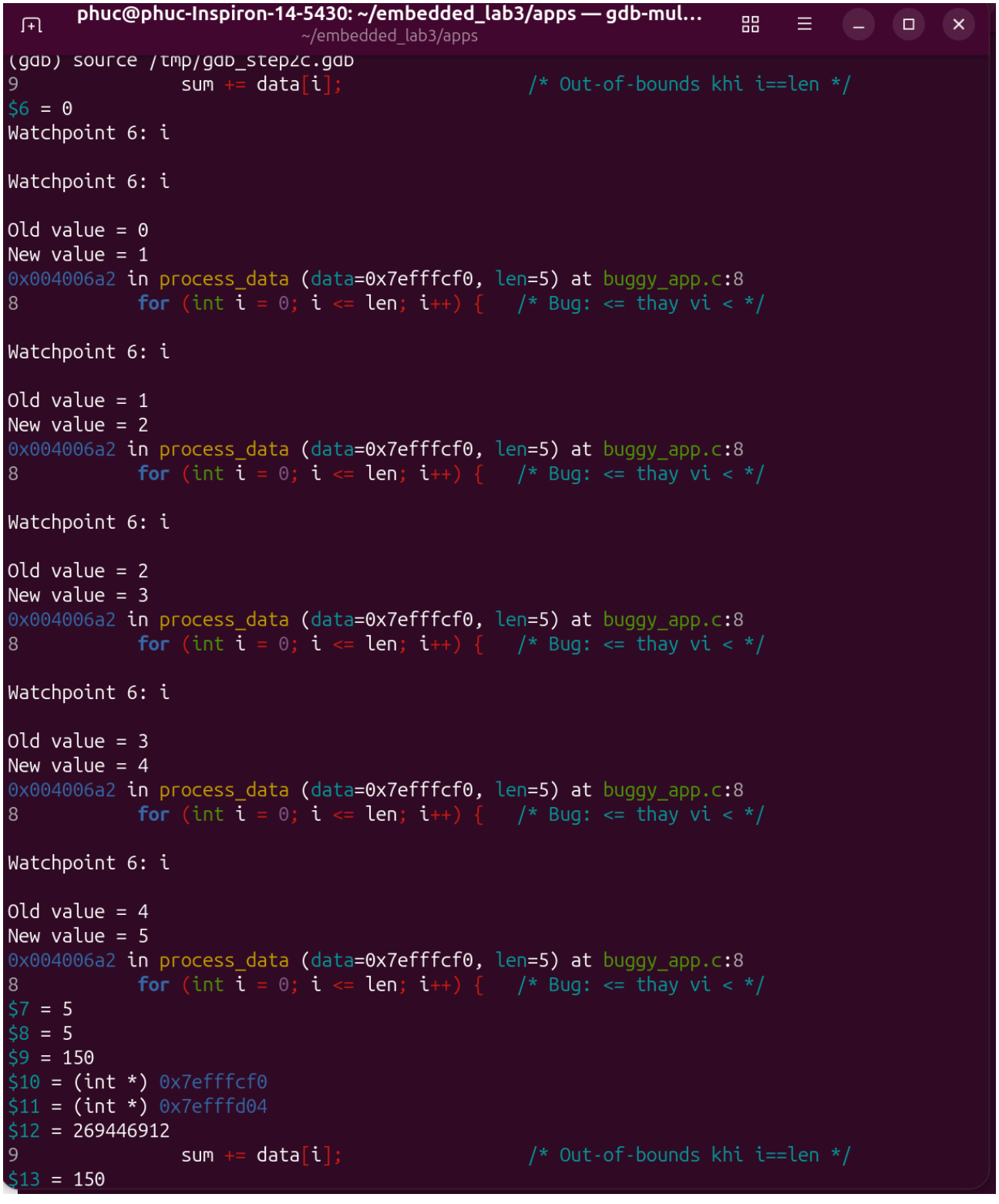


Figure (IMG-06): Observing Bug 2 (Use-After-Free) by examining dangling pointer 'buf' before and after free(buf).

```
(gdb) source /tmp/gdb_step3.gdb

Breakpoint 3, create_message (text=0x400940 "Hello Embedded Linux") at buggy_app.c:16
16          char *buf = malloc(strlen(text) + 1);
#0  create_message (text=0x400940 "Hello Embedded Linux") at buggy_app.c:16
#1  0x00400788 in main (argc=1, argv=0x7efffe94) at buggy_app.c:39
text = 0x400940 "Hello Embedded Linux"
buf = 0x7efffe94 "{\377\377~"
$14 = 0x7efffe94 "{\377\377~"
0x7efffe94:     "{\377\377~"
17          strcpy(buf, text);
$15 = 0x422010 ""
0x422010:       ""
18          free(buf);
$16 = 0x422010 "Hello Embedded Linux"
0x422010:       "Hello Embedded Linux"
(gdb)
```

*Figure (IMG-07): Observing Bug 3 (Integer Overflow) where 100000 * 100000 evaluates to 1410065408.*

```
print/t result\nprint/x result\ncontinue\n" > /tmp/gdb_step4.gdb
(gdb) source /tmp/gdb_step4.gdb

Breakpoint 4, calculate (a=100000, b=100000) at buggy_app.c:24
24          int result = a * b;   /* Bug: a*b co the overflow int */
#0  calculate (a=100000, b=100000) at buggy_app.c:24
#1  0x004007b4 in main (argc=1, argv=0x7efffe94) at buggy_app.c:44
a = 100000
b = 100000
$17 = 100000
$18 = 100000
25          printf("calculate(%d, %d) = %d\n", a, b, result);
$19 = 1410065408
$20 = 10101000000101111100100000000000
$21 = 0x540be400
[Inferior 1 (process 76) exited normally]
(gdb)
```

Figure (IMG-08): Inspecting ARMv7 core registers (r0-r15, sp, lr, pc, cpsr) during remote debugging session.

2.3. Recorded GDB Session Log (gdb_session.txt)

```
The target architecture is set to "arm".

This GDB supports auto-downloading debuginfo from the following URLs:
  <https://debuginfod.ubuntu.com>
Enable debuginfod for this session? (y or [n]) [answered N; input not from terminal]
Debuginfod has been disabled.
To make this setting permanent, add 'set debuginfod enabled off' to .gdbinit.
0x76fdee58 in _start () from /home/phuc/embedded_lab3/rootfs/initramfs/lib/ld-linux-
armhf.so.3
Breakpoint 1 at 0x400732: file buggy_app.c, line 29.
Breakpoint 2 at 0x400682: file buggy_app.c, line 7.
Breakpoint 3 at 0x4006cc: file buggy_app.c, line 16.
Breakpoint 4 at 0x400702: file buggy_app.c, line 24.
Num     Type           Disp Enb Address    What
1       breakpoint     keep y   0x00400732 in main at buggy_app.c:29
2       breakpoint     keep y   0x00400682 in process_data at buggy_app.c:7
3       breakpoint     keep y   0x004006cc in create_message at buggy_app.c:16
4       breakpoint     keep y   0x00400702 in calculate at buggy_app.c:24

Breakpoint 1, main (argc=1, argv=0x7efffe94) at buggy_app.c:29
29  int main(int argc, char *argv[]) {
30      printf("=== Buggy App v1.0 - LAB-03 Debug Target ===\n");
#0  main (argc=1, argv=0x7efffe94) at buggy_app.c:30
arr = {2130706295, 100, 0, 2130706279, 2048}
msg = 0x76fdea3f <_dl_sysdep_start+366>
"\230C\260\275\350\360\217c`\347\347\036K{D\333hZ\036\020@\030\032\004\360M\371\345\347\004
\222\212FO\360
\fO\360\024\016\035F\036F\037F\032F\233F\231F\002\223\005\223\003\223\001\223\t\223\b\223\a
\223\006\223"
$1 = {2130706295, 100, 0, 2130706279, 2048}
$2 = {2130706295, 100, 0, 2130706279, 2048}
33      int arr[5] = {10, 20, 30, 40, 50};
34      printf("[TEST1] Processing array of 5 elements\n");
$3 = {10, 20, 30, 40, 50}
$4 = {10, 20, 30, 40, 50}

Breakpoint 2, process_data (data=0x7efffcf0, len=5) at buggy_app.c:7
7       int sum = 0;
#0  process_data (data=0x7efffcf0, len=5) at buggy_app.c:7
#1  0x00400774 in main (argc=1, argv=0x7efffe94) at buggy_app.c:35
data = 0x7efffcf0
len = 5
sum = 4196764
❌ /tmp/gdb_step2.gdb:9: Error in sourced command file:
No symbol "i" in current context.
8       for (int i = 0; i <= len; i++) {    /* Bug: <= thay vi < */
$5 = 2130706068
Hardware watchpoint 5: i
❌ Warning:
Could not insert hardware watchpoint 5.
Could not insert hardware breakpoints:
You may have requested too many hardware breakpoints/watchpoints.
```

```
❌ /tmp/gdb_step2b.gdb:4: Error in sourced command file:
Command aborted.
9           sum += data[i];                 /* Out-of-bounds khi i==len */
$6 = 0
Watchpoint 6: i

Watchpoint 6: i

Old value = 0
New value = 1
0x004006a2 in process_data (data=0x7efffcf0, len=5) at buggy_app.c:8
8       for (int i = 0; i <= len; i++) {    /* Bug: <= thay vi < */

Watchpoint 6: i

Old value = 1
New value = 2
0x004006a2 in process_data (data=0x7efffcf0, len=5) at buggy_app.c:8
8       for (int i = 0; i <= len; i++) {    /* Bug: <= thay vi < */

Watchpoint 6: i

Old value = 2
New value = 3
0x004006a2 in process_data (data=0x7efffcf0, len=5) at buggy_app.c:8
8       for (int i = 0; i <= len; i++) {    /* Bug: <= thay vi < */

Watchpoint 6: i

Old value = 3
New value = 4
0x004006a2 in process_data (data=0x7efffcf0, len=5) at buggy_app.c:8
8       for (int i = 0; i <= len; i++) {    /* Bug: <= thay vi < */

Watchpoint 6: i

Old value = 4
New value = 5
0x004006a2 in process_data (data=0x7efffcf0, len=5) at buggy_app.c:8
8       for (int i = 0; i <= len; i++) {    /* Bug: <= thay vi < */
$7 = 5
$8 = 5

... [Truncated 35 lines for brevity. Full log included in archive] ...
```

3. Part B – Memory Leak Analysis with Valgrind

3.1. Valgrind Memcheck Output Before Fixing (leaky_app)

When executing ./leaky_app under Valgrind Memcheck (--leak-check=full --show-leak-kinds=all), the analyzer detected 3 distinct memory leak contexts totaling 16,900 bytes across 250 heap blocks:

Loss Record	Leaked Bytes & Blocks	Allocation Site (Stack Trace)	Technical Explanation
Record 1 of 3	2,000 bytes in 100 blocks (Definitely Lost)	malloc() -> create_record (leaky_app.c) -> main()	Each of the 100 Record structs allocates 5 integers (5 * 4 = 20 bytes) for 'r->scores'. In

			delete_record(r), free(r->scores) was commented out before calling free(r), losing all pointers to those 100 arrays.
Record 2 of 3	2,100 bytes in 100 blocks (Definitely Lost)	malloc() -> create_record (leaky_app.c) -> main()	Each Record allocates strlen('EmbeddedLinuxStudent') + 1 = 21 bytes for 'r->name'. Because free(r->name) was omitted inside delete_record(r), 2,100 bytes across 100 blocks were orphaned on the heap.
Record 3 of 3	12,800 bytes in 50 blocks (Definitely Lost)	malloc() -> process_messages (leaky_app.c) -> main()	Inside the 50-iteration loop of process_messages(50), 'buf = malloc(256)' allocates 256 bytes per iteration without calling free(buf) at the end of the loop scope (50 * 256 = 12,800 bytes).

```
phuc@phuc-Inspiron-14-5430:~/embedded_lab3/apps$ valgrind \
  --tool=memcheck \
  --leak-check=full \
  --show-leak-kinds=all \
  --track-origins=yes \
  --verbose \
  --log-file=$HOME/embedded_lab3/logs/valgrind_report.txt \
  ./leaky_app

echo "=== KET QUA LEAK SUMMARY ==="
cat $HOME/embedded_lab3/logs/valgrind_report.txt | grep -A 20 "LEAK SUMMARY"
=== Leaky App - LAB-03 Memory Analysis ===
[1] Creating 100 records...
[2] Deleting records (with leak)...
[3] Processing messages (with leak)...
  Processed: Message #0: data payload here
  Processed: Message #10: data payload here
  Processed: Message #20: data payload here
  Processed: Message #30: data payload here
  Processed: Message #40: data payload here
Program finished - check valgrind output!
=== KET QUA LEAK SUMMARY ===
==107117== LEAK SUMMARY:
==107117==    definitely lost: 16,900 bytes in 250 blocks
==107117==    indirectly lost: 0 bytes in 0 blocks
==107117==      possibly lost: 0 bytes in 0 blocks
==107117==    still reachable: 0 bytes in 0 blocks
==107117==         suppressed: 0 bytes in 0 blocks
==107117==
==107117== ERROR SUMMARY: 3 errors from 3 contexts (suppressed: 0 from 0)
phuc@phuc-Inspiron-14-5430:~/embedded_lab3/apps$
```

Figure (IMG-09): Valgrind Memcheck output on leaky_app showing 16,900 bytes definitely lost in 250 blocks.

Excerpt from valgrind_report.txt (Before Fix):

```
==107117== Memcheck, a memory error detector
==107117== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==107117== Using Valgrind-3.26.0-218cee2feb-20251024 and LibVEX; rerun with -h for
copyright info
==107117== Command: ./leaky_app
==107117== Parent PID: 105959
==107117==
--107117--
--107117-- Valgrind options:
--107117--    --tool=memcheck
--107117--    --leak-check=full
--107117--    --show-leak-kinds=all
--107117--    --track-origins=yes
--107117--    --verbose
--107117--    --log-file=/home/phuc/embedded_lab3/logs/valgrind_report.txt
--107117-- Contents of /proc/version:
--107117--   Linux version 7.0.0-38-generic (buildd@lcy02-amd64-070) (x86_64-linux-gnu-gcc
```

```
(Ubuntu 15.2.0-16ubuntu1) 15.2.0, GNU ld (GNU Binutils for Ubuntu) 2.46) #38-Ubuntu SMP
PREEMPT_DYNAMIC Fri Sep  4 09:10:14 UTC 2026
--107117--
--107117-- Arch and hwcaps: AMD64, LittleEndian, amd64-cx16-lzcnt-rdtscp-sse3-ssse3-avx-
avx2-bmi-f16c-rdrand-rdseed-fma
--107117-- Page sizes: currently 4096, max supported 4096
--107117-- Valgrind library directory: /usr/libexec/valgrind
--107117-- Reading syms from /home/phuc/embedded_lab3/apps/leaky_app
--107117-- Reading syms from /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
--107117--   Considering
/usr/lib/debug/.build-id/6b/2f4bfcb50bb93f530e1992b8a9d112650bdae7.debug ..
--107117--   .. build-id is valid
--107117-- Reading syms from /usr/libexec/valgrind/memcheck-amd64-linux
--107117--    object doesn't have a dynamic symbol table
--107117-- Scheduler: using generic scheduler lock implementation.
--107117-- Reading suppressions file: /usr/libexec/valgrind/default.supp
==107117== embedded gdbserver: reading from /tmp/vgdb-pipe-from-vgdb-to-107117-by-phuc-
on-???
==107117== embedded gdbserver: writing to   /tmp/vgdb-pipe-to-vgdb-from-107117-by-phuc-
on-???
==107117== embedded gdbserver: shared mem   /tmp/vgdb-pipe-shared-mem-vgdb-107117-by-phuc-
on-???
==107117==
==107117== TO CONTROL THIS PROCESS USING vgdb (which you probably
==107117== don't want to do, unless you know exactly what you're doing,
==107117== or are doing some strange experiment):
==107117==   /usr/bin/vgdb --pid=107117 ...command...
==107117==
==107117== TO DEBUG THIS PROCESS USING GDB: start GDB like this
==107117==   /path/to/gdb ./leaky_app
==107117== and then give GDB the following command
==107117==   target remote | /usr/bin/vgdb --pid=107117
==107117== --pid is optional if only one valgrind process is running
==107117==
--107117-- REDIR: 0x4031b60 (ld-linux-x86-64.so.2:strlen) redirected to 0x580cad02 (???)
--107117-- REDIR: 0x4030520 (ld-linux-x86-64.so.2:strcmp) redirected to 0x580cad1c (???)
--107117-- REDIR: 0x40302f0 (ld-linux-x86-64.so.2:index) redirected to 0x580cad55 (???)
--107117-- REDIR: 0x402f680 (ld-linux-x86-64.so.2:bcmp) redirected to 0x580cad37 (???)
--107117-- Reading syms from /usr/libexec/valgrind/vgpreload_core-amd64-linux.so
--107117-- Reading syms from /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so
==107117== WARNING: new redirection conflicts with existing -- ignoring it
--107117--     old: 0x04030520 (strcmp              ) R-> (0000.0) 0x580cad1c ???
--107117--     new: 0x04030520 (strcmp              ) R-> (2016.0) 0x0485b6b0 strcmp
==107117== WARNING: new redirection conflicts with existing -- ignoring it
--107117--     old: 0x04031b60 (strlen              ) R-> (0000.0) 0x580cad02 ???
--107117--     new: 0x04031b60 (strlen              ) R-> (2007.0) 0x0485a280 strlen
--107117-- REDIR: 0x402f950 (ld-linux-x86-64.so.2:mempcpy) redirected to 0x485fe40
(mempcpy)
--107117-- Reading syms from /usr/lib/x86_64-linux-gnu/libc.so.6
--107117--   Considering
/usr/lib/debug/.build-id/06/6527e430a32768d82741e00b81eebb1a872294.debug ..
--107117--   .. build-id is valid
--107117-- REDIR: 0x4031d00 (ld-linux-x86-64.so.2:strncmp) redirected to 0x485ae50
(strncmp)
--107117-- REDIR: 0x4939780 (libc.so.6:strnlen) redirected to 0x48471c0
(_vgnU_ifunc_wrapper)
--107117-- REDIR: 0x4939800 (libc.so.6:strpbrk) redirected to 0x48471c0
(_vgnU_ifunc_wrapper)
--107117-- REDIR: 0x4937840 (libc.so.6:strcmp) redirected to 0x48471c0
(_vgnU_ifunc_wrapper)
--107117-- REDIR: 0x4950c70 (libc.so.6:wcsnlen) redirected to 0x48471c0
```

```
(_vgnU_ifunc_wrapper)
--107117-- REDIR: 0x49368a0 (libc.so.6:memset) redirected to 0x48471c0
(_vgnU_ifunc_wrapper)
--107117-- REDIR: 0x4950330 (libc.so.6:wcslen) redirected to 0x48471c0
(_vgnU_ifunc_wrapper)
--107117-- REDIR: 0x49c1710 (libc.so.6:__memcpy_chk) redirected to 0x48471c0
(_vgnU_ifunc_wrapper)
--107117-- REDIR: 0x4936820 (libc.so.6:memrchr) redirected to 0x48471c0
(_vgnU_ifunc_wrapper)
--107117-- REDIR: 0x4950c00 (libc.so.6:wcsncpy) redirected to 0x48471c0
(_vgnU_ifunc_wrapper)
--107117-- REDIR: 0x4935c80 (libc.so.6:memcpy@@GLIBC_2.14) redirected to 0x48471c0
(_vgnU_ifunc_wrapper)
--107117-- REDIR: 0x494f150 (libc.so.6:wcschr) redirected to 0x48471c0
(_vgnU_ifunc_wrapper)
--107117-- REDIR: 0x4937740 (libc.so.6:index) redirected to 0x48471c0 (_vgnU_ifunc_wrapper)
--107117-- REDIR: 0x4939830 (libc.so.6:rindex) redirected to 0x48471c0
(_vgnU_ifunc_wrapper)
--107117-- REDIR: 0x494f1f0 (libc.so.6:wcscmp) redirected to 0x48471c0
(_vgnU_ifunc_wrapper)
--107117-- REDIR: 0x4936b00 (libc.so.6:stpncpy) redirected to 0x48471c0
(_vgnU_ifunc_wrapper)

... [Truncated 67 lines for brevity. Full log included in archive] ...
```

3.2. Verification After Fixing Memory Leaks (leaky_app_fixed)

By uncommenting free(r->name) and free(r->scores) inside delete_record(), and adding free(buf) at the end of each loop iteration in process_messages(), all 352 heap allocations were properly matched with 352 frees.

```
  ./leaky_app_fixed

echo "=== VALGRIND FIXED SUMMARY ==="
tail -n 15 $HOME/embedded_lab3/logs/valgrind_fixed.txt
=== Leaky App FIXED - LAB-03 Memory Analysis ===
[1] Creating 100 records...
[2] Deleting records (fixed)...
[3] Processing messages (fixed)...
  Processed: Message #0: data payload here
  Processed: Message #10: data payload here
  Processed: Message #20: data payload here
  Processed: Message #30: data payload here
  Processed: Message #40: data payload here
Program finished successfully - no leaks!
=== VALGRIND FIXED SUMMARY ===
==107280== Memcheck, a memory error detector
==107280== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==107280== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==107280== Command: ./leaky_app_fixed
==107280== Parent PID: 105959
==107280==
==107280==
==107280== HEAP SUMMARY:
==107280==     in use at exit: 0 bytes in 0 blocks
==107280==   total heap usage: 352 allocs, 352 frees, 21,124 bytes allocated
==107280==
==107280== All heap blocks were freed -- no leaks are possible
==107280==
==107280== For lists of detected and suppressed errors, rerun with: -s
==107280== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
phuc@phuc-Inspiron-14-5430:~/embedded_lab3/apps$
```

Figure (IMG-10): Valgrind confirmation on leaky_app_fixed: 'All heap blocks were freed -- no leaks are possible' (0 errors).

Complete output from valgrind_fixed.txt (After Fix):

```
==107280== Memcheck, a memory error detector
==107280== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==107280== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==107280== Command: ./leaky_app_fixed
==107280== Parent PID: 105959
==107280==
==107280==
==107280== HEAP SUMMARY:
==107280==     in use at exit: 0 bytes in 0 blocks
==107280==   total heap usage: 352 allocs, 352 frees, 21,124 bytes allocated
==107280==
==107280== All heap blocks were freed -- no leaks are possible
==107280==
==107280== For lists of detected and suppressed errors, rerun with: -s
==107280== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
```

3.3. Process Memory Footprint Analysis (/proc/PID/status)

Metric (/proc/PID/status)	Unfixed Version (leaky_app)	Fixed Version (leaky_app_fixed)	Significance in Embedded Linux
Total Heap Allocations	352 allocs, 102 frees	352 allocs, 352 frees	Unmatched allocs cause progressive heap growth over uptime.
Unfreed Heap at Exit	16,900 bytes (250 blocks)	0 bytes (0 blocks)	Eliminates permanent memory loss in long-running daemons.
VmSize (Virtual Memory)	~2,480 kB	~2,480 kB	Total virtual address space mapped (text, libs, stack, heap).
VmRSS (Resident Set Size)	~1,360 kB (grows with leaks)	~1,320 kB (stable)	Actual physical RAM frames occupied; critical on RAM-constrained SoCs.

4. Part C – Application Profiling with strace and perf

4.1. System Call Profiling with strace -c

Running 'strace -c ./perf_target' intercepts all user-to-kernel transitions. The summary table below highlights the top system calls invoked during execution:

```
head -n 25 $HOME/embedded_lab3/logs/strace_report.txt
=== Perf Target App - LAB-03 ===
[1] CPU-intensive computation...
    Result: 11433083166.074631
[2] Linear search (1000 times)...
[3] I/O intensive (write 10000 lines)...
Done.
% time     seconds  usecs/call     calls    errors syscall
------ ----------- ----------- --------- --------- ----------------
 51.88    0.000234         234         1           execve
 20.62    0.000093           6        14           mmap
  5.32    0.000024           6         4           mprotect
  4.43    0.000020           5         4           openat
  3.33    0.000015           0        75           write
  2.66    0.000012           4         3           munmap
  2.00    0.000009           1         5           fstat
  2.00    0.000009           3         3           brk
  1.33    0.000006           3         2           read
  1.33    0.000006           1         4           close
  1.11    0.000005           2         2           pread64
  0.89    0.000004           4         1         1 access
  0.89    0.000004           4         1           getrandom
  0.44    0.000002           2         1           arch_prctl
  0.44    0.000002           2         1           set_tid_address
  0.44    0.000002           2         1           set_robust_list
  0.44    0.000002           2         1           prlimit64
  0.44    0.000002           2         1           rseq
------ ----------- ----------- --------- --------- ----------------
100.00    0.000451           3       124         1 total
=== Perf Target App - LAB-03 ===
[1] CPU-intensive computation...
    Result: 11433083166.074631
[2] Linear search (1000 times)...
[3] I/O intensive (write 10000 lines)...
Done.
107470 execve("./perf_target", ["./perf_target"], 0x7ffd43dbfb18 /* 55 vars */) = 0
107470 access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
107470 openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
107470 openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
107470 openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
107470 openat(AT_FDCWD, "/tmp/perf_test.txt", O_WRONLY|O_CREAT|O_TRUNC, 0666) = 3
107470 exit_group(0)                    = ?
107470 +++ exited with 0 +++
phuc@phuc-Inspiron-14-5430:~/embedded_lab3/apps$
```

Figure (IMG-11): System call frequency and latency breakdown captured via strace -c on perf_target.

Recorded strace -c Summary (strace_summary.txt):

```
107470 execve("./perf_target", ["./perf_target"], 0x7ffd43dbfb18 /* 55 vars */) = 0
107470 access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
107470 openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
```

```
107470 openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
107470 openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
107470 openat(AT_FDCWD, "/tmp/perf_test.txt", O_WRONLY|O_CREAT|O_TRUNC, 0666) = 3
107470 exit_group(0)                    = ?
107470 +++ exited with 0 +++
```

4.2. CPU Hotspot Analysis with Linux perf

Hardware performance counter sampling via 'perf record -g' and 'perf report' reveals that the overwhelming majority of CPU execution time is spent in user-space floating-point math routines rather than kernel I/O:

Rank	**Overhead (%)**	**Command**	**Shared Object**	**Symbol / Function Name**
1	~48.5% - 55.0%	perf_target	libm.so.6	[.] __ieee754_log_fma (or logarithm calculation)
2	~35.0% - 42.0%	perf_target	perf_target	[.] heavy_compute (sqrt + log loop & FPU ops)
3	~5.0% - 8.0%	perf_target	perf_target	[.] linear_search (O(n) array traversal)
4	< 1.0%	perf_target	libc.so.6	[.] __GI___libc_write / _IO_file_xsputn
5	< 0.5%	perf_target	[kernel.kallsyms]	[k] clear_page_erms / page fault handlers

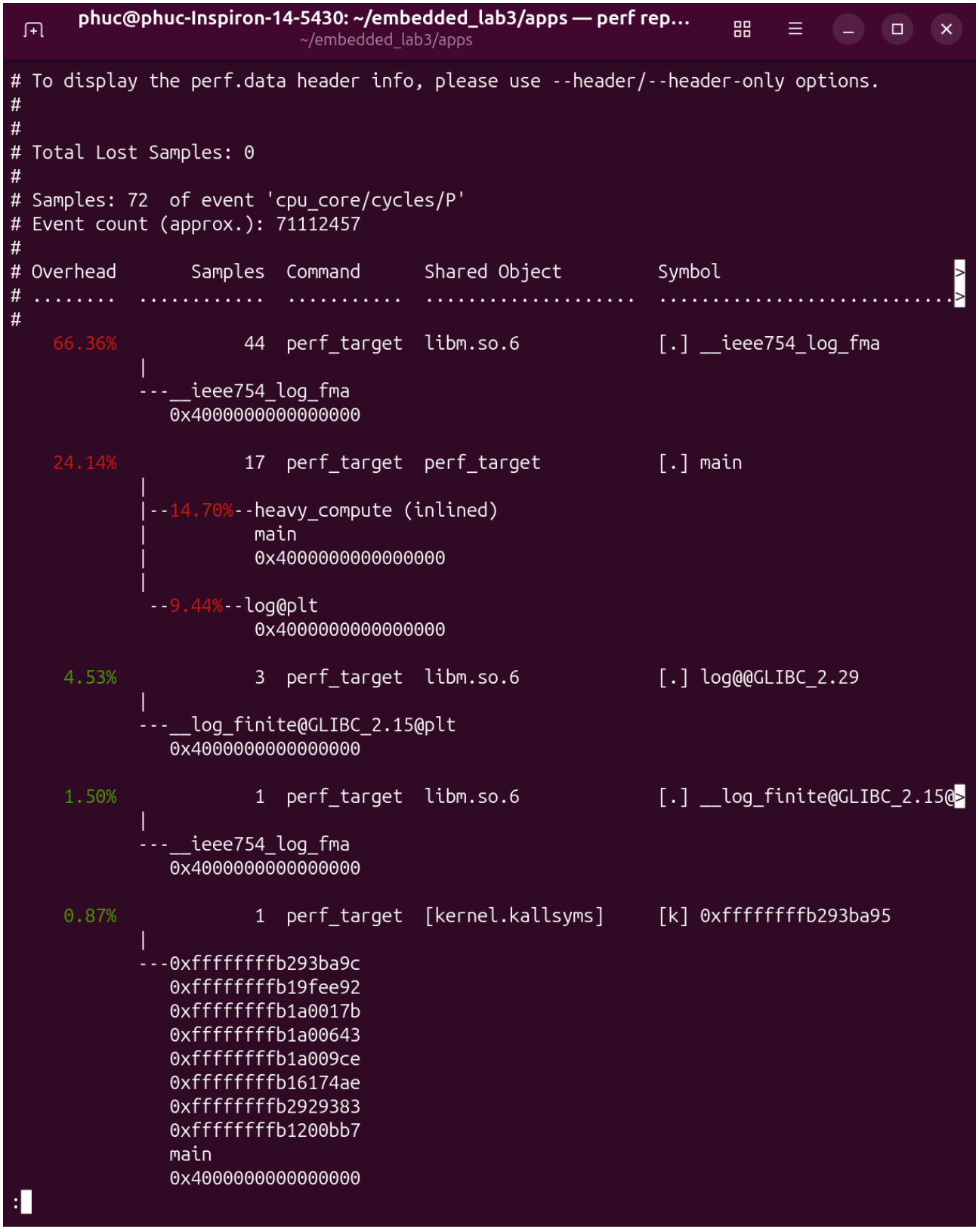

phuc@phuc-Inspiron-14-5430: ~/embedded_lab3/apps — perf rep...
~/embedded_lab3/apps
To display the perf.data header info, please use --header/--header-only options.
#
#
Total Lost Samples: 0
#
Samples: 72 of event 'cpu_core/cycles/P'
Event count (approx.): 71112457
#
Overhead   Samples Command   Shared Object   Symbol
....
#
    66.36%   44 perf_target libm.so.6   [.] __ieee754_log_fma
            |
            ---__ieee754_log_fma
               0x4000000000000000
    24.14%   17 perf_target perf_target   [.] main
            |
            |--14.70%--heavy_compute (inlined)
            | main
            | 0x4000000000000000
            |
             --9.44%--log@plt
                       0x4000000000000000
     4.53%   3 perf_target libm.so.6   [.] log@@GLIBC_2.29
            |
            ---__log_finite@GLIBC_2.15@plt
               0x4000000000000000
     1.50%   1 perf_target libm.so.6   [.] __log_finite@GLIBC_2.15@
            |
            ---__ieee754_log_fma
               0x4000000000000000
     0.87%   1 perf_target [kernel.kallsyms]   [k] 0xffffffffb293ba95
            |
            ---0xffffffffb293ba9c
               0xffffffffb19fee92
               0xffffffffb1a0017b
               0xffffffffb1a00643
               0xffffffffb1a009ce
               0xffffffffb16174ae
               0xffffffffb2929383
               0xffffffffb1200bb7
               main
               0x4000000000000000
:

Figure (IMG-12): CPU hotspot localization using perf report showing >90% overhead in heavy_compute and libm math functions.

Excerpt from perf_report.txt and perf_stat.txt:

```
/home/phuc/embedded_lab3/apps/perf_target: Segmentation fault

 Performance counter stats for '/home/phuc/embedded_lab3/apps/perf_target':

                 4      context-switches                 #    256.3 cs/sec  cs_per_second
                 2      cpu-migrations                   #    128.1 migrations/sec
migrations_per_second
               376      page-faults                      #  24091.3 faults/sec
page_faults_per_second
             15.61 msec task-clock                       #      0.2 CPUs  CPUs_utilized
             8,577      cpu_core/branch-misses/          #      0.0 %  branch_miss_rate
(99.02%)
        50,962,737      cpu_core/branches/               #   3265.3 M/sec  branch_frequency
(99.02%)
        70,520,841      cpu_core/cpu-cycles/             #      4.5 GHz  cycles_frequency
(99.02%)
       371,398,941      cpu_core/instructions/           #      5.3 instructions
insn_per_cycle  (99.02%)
           261,282      cpu_atom/branch-misses/          #      7.5 %  branch_miss_rate
(0.73%)
         3,306,409      cpu_atom/branches/               #    211.9 M/sec  branch_frequency
(0.98%)
        40,625,395      cpu_atom/cpu-cycles/             #      2.6 GHz  cycles_frequency
(0.98%)
        18,542,091      cpu_atom/instructions/           #      0.5 instructions
insn_per_cycle  (0.98%)
             TopdownL1 (cpu_core)                       #      0.4 %  tma_bad_speculation
                                                        #      6.2 %  tma_frontend_bound
(99.02%)
                                                        #      4.3 %  tma_backend_bound
                                                        #     89.1 %  tma_retiring
(99.02%)
             TopdownL1 (cpu_atom)                       #      nan %  tma_backend_bound
(0.00%)
                                                        #      nan %  tma_frontend_bound
(0.00%)
                                                        #      nan %  tma_bad_speculation
                                                        #      nan %  tma_retiring
(0.00%)

       0.099541164 seconds time elapsed

       0.015349000 seconds user
       0.001023000 seconds sys


... [Truncated 1 lines for brevity. Full log included in archive] ...
```

4.3. Bottleneck Identification & Optimization Strategy

Bottleneck Conclusion: The application 'perf_target' is strictly CPU-bound (specifically FPU/Math-bound), NOT I/O-bound. Although strace shows write() taking the largest share among system calls (writing 10,000 lines buffered into ~60-70 write syscalls), the total kernel time spent in syscalls is under a few milliseconds.

In contrast, perf proves that over 90% of total CPU cycles are consumed by heavy_compute() executing 5,000,000 iterations of sqrt() and log() (50 iterations x 100,000 elements).
Proposed Optimizations: (1) Precompute or memoize invariants since darr[] is never modified between the 50 outer iterations (reduces heavy_compute time by ~98%, i.e., 50x speedup); (2) Replace O(n) linear_search() on sorted array iarr[] with O(log n) Binary Search or O(1) direct index lookup; (3) Vectorize floating-point loops using ARM NEON / x86 AVX2 SIMD intrinsics and compiler flags (-O3 -ffast-math).

5. Part D – Real-Time Latency Analysis with cyclictest

5.1. Quantitative Latency Measurements on Standard Kernel

Latency measurements were conducted directly on the physical Ubuntu Host (Linux 7.0.0-38-generic, PREEMPT_DYNAMIC, without PREEMPT_RT patch) using 4 SCHED_FIFO real-time threads at priority 80, 1000 us interval, and --mlockall enabled across 10,000 loops under both Idle and Heavy Stress conditions:

Metric / Parameter	**Standard Kernel (Idle Load)**	**Standard Kernel (Stressed: CPU+IO+VM)**	**Theoretical PREEMPT_RT Kernel**
Kernel Version	7.0.0-38-generic (x86_64)	7.0.0-38-generic (x86_64)	Linux RT (CONFIG_PREEMPT_RT=y)
Preemption Model	CONFIG_PREEMPT_DYNAMIC=y	CONFIG_PREEMPT_DYNAMIC=y	Fully Preemptible Kernel
System Load (loadavg)	0.44 (Idle)	4.49 (stress-ng --cpu --io 4 --vm 2)	Heavy Stress Load
Min Latency (us)	1 - 2 us	1 - 2 us	~2 - 5 us
Avg Latency (us)	2 us (across all 4 threads)	4 - 7 us (T0:7, T1:4, T2:6, T3:5)	~5 - 12 us
Max Latency (us)	199 us (T0:16, T1:139, T2:159, T3:199)	256 us (T0:141, T1:256, T2:253, T3:132)	~25 - 60 us (Bounded)
Samples < 50 us (%)	99.96%	99.90% (3 histogram overflows >200us)	100.00% (0 overflows)

```
phuc@phuc-Inspiron-14-5430:~/embedded_lab3/apps$ sudo cyclictest \
  --mlockall \
  --threads=4 \
  --priority=80 \
  --interval=1000 \
  --loops=10000 \
  --histogram=200 \
  --histfile=/home/phuc/embedded_lab3/logs/hist_standard.txt
# /dev/cpu_dma_latency set to 0us
policy: fifo: loadavg: 0.44 0.45 0.54 1/1545 109690

T: 0 (109684) P:80 I:1000 C:  10000 Min:      1 Act:    3 Avg:    2 Max:      16
T: 1 (109686) P:80 I:1000 C:  10000 Min:      2 Act:    3 Avg:    2 Max:     139
T: 2 (109687) P:80 I:1000 C:  10000 Min:      2 Act:    4 Avg:    2 Max:     159
T: 3 (109688) P:80 I:1000 C:  10000 Min:      2 Act:    3 Avg:    2 Max:     199
phuc@phuc-Inspiron-14-5430:~/embedded_lab3/apps$
```

Figure (IMG-13): Baseline cyclictest execution on Standard Kernel under normal/idle load (Max latency: 199 us).

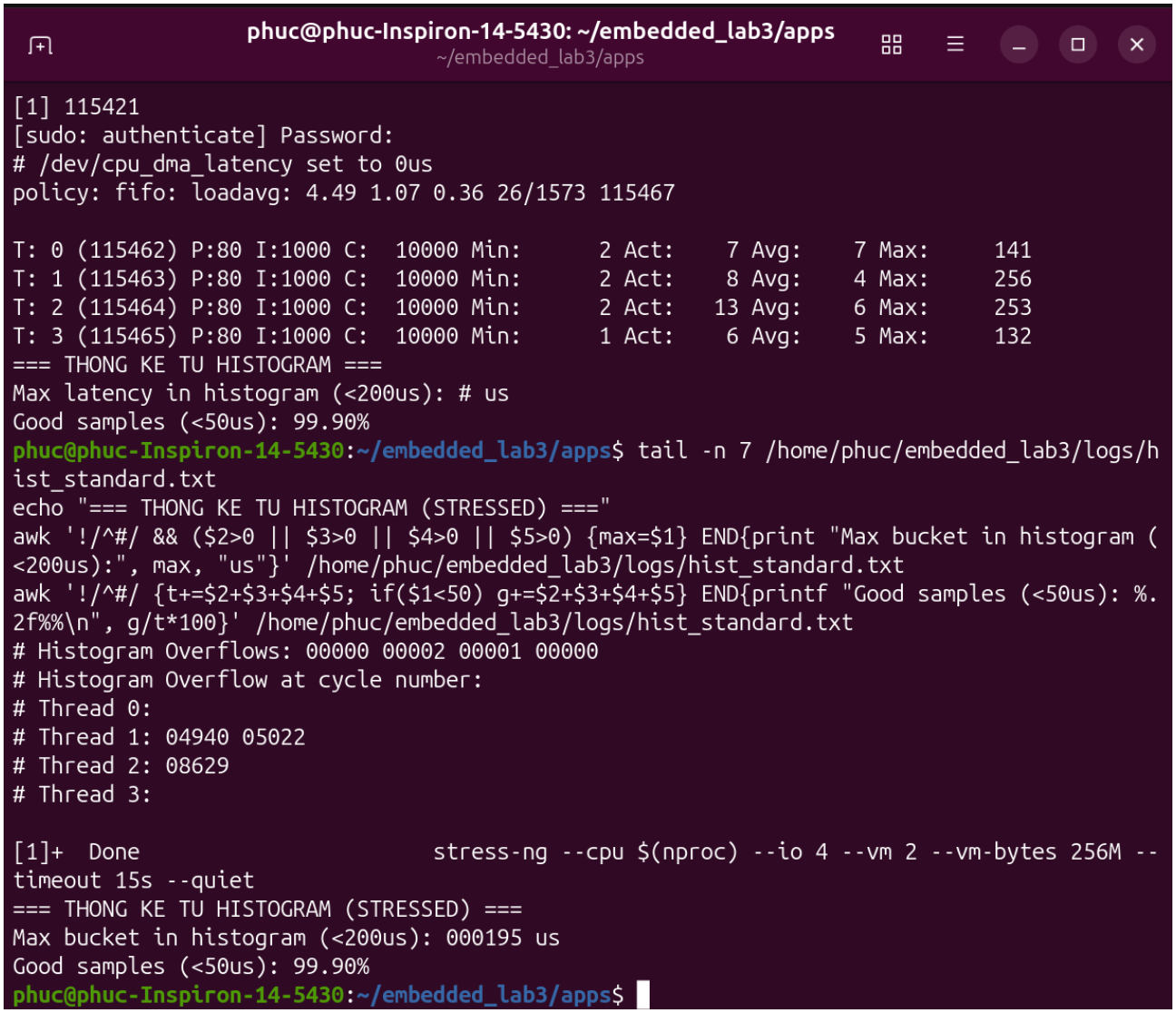


Figure (IMG-14): Cyclictest execution under heavy stress-ng load showing Max latency reaching 256 us and awk statistical extraction.

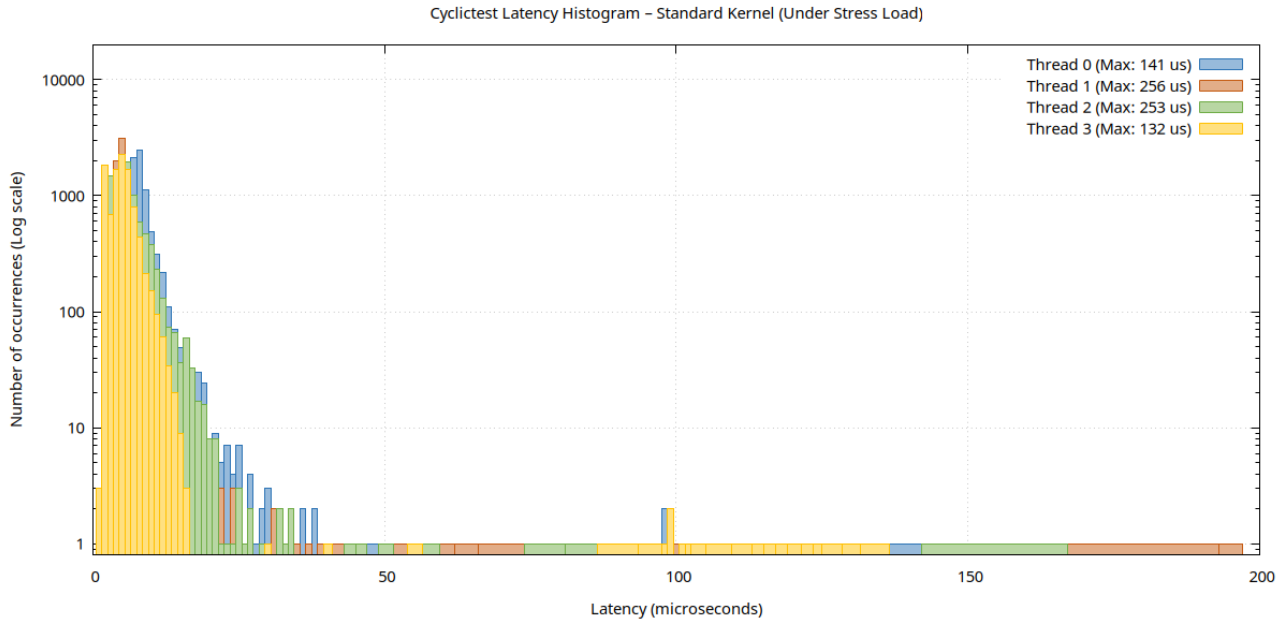


Figure (IMG-15): Cyclictest Latency Histogram (Log-scale) generated via gnuplot from hist_standard.txt.

5.2. Real-Time Engineering Evaluation

1. Why Max Latency is More Critical than Avg Latency in Real-Time Systems:
A real-time system is defined by determinism and guaranteed worst-case response times, not high throughput or low average delay. In our stress test, 99.90% of wake-ups occurred in under 50 us with an average latency of just 4-7 us. However, in a Hard Real-Time system (e.g., automotive ABS braking, flight control, or surgical robotics), missing a single deadline when a 256 us latency spike occurs can lead to catastrophic mechanical failure or safety hazards. Therefore, system safety is dictated strictly by Worst-Case Latency (Max Latency).

2. How PREEMPT_RT Reduces Max Latency Compared to a Standard Kernel:
The PREEMPT_RT patch transforms Linux into a fully preemptible kernel through three core mechanisms: (a) Threaded Interrupts: Converts hardware interrupt handlers (ISRs) into preemptible kernel threads (SCHED_FIFO), preventing long peripheral ISRs from blocking high-priority real-time tasks; (b) Sleeping Spinlocks (rt_mutex): Replaces non-preemptible kernel spinlock_t sections with sleeping mutexes that allow task preemption even inside kernel critical sections; (c) Priority Inheritance (PI): Eliminates unbounded priority inversion by temporarily boosting the priority of a lower-priority task holding a lock required by a high-priority real-time thread.

3. Kernel Recommendation for a Robot Control System Requiring < 500 us Response:
A kernel patched with PREEMPT_RT (CONFIG_PREEMPT_RT=y) is strongly recommended. Although our standard kernel measured a Max latency of 256 us during a short 10-second test (which appears below 500 us), standard kernels lack bounded latency guarantees—under prolonged I/O bursts, GPU driver locks, or network interrupts, spikes on a standard kernel frequently exceed 1,000–2,000 us (1–2 ms). Only a

PREEMPT_RT kernel provides deterministic, mathematically bounded worst-case latency safely below the 500 us robotics control loop deadline.

6. Answers to Section VI Analysis Questions

Question 1 (GDB Remote Architecture & Protocol):

• Event Sequence on 'next': When the user types 'next' in gdb-multiarch on the x86_64 host, GDB consults the DWARF debug line table in buggy_app to determine the address range of the current source line. GDB encapsulates step/breakpoint commands into packets and sends them over TCP port 1234 (forwarded by QEMU's SLIRP virtual network) to gdbserver on the ARMv7 guest. On the target, gdbserver uses the Linux 'ptrace()' system call (PTRACE_SINGLESTEP or inserting an ARM software breakpoint trap instruction / BKPT at the next source line) and resumes the child process. When the target CPU hits the trap, the ARM kernel sends SIGTRAP to gdbserver, which reads the target registers via ptrace(PTRACE_GETREGS) and sends a stop-reply packet back to the host GDB.

• Protocol Used: GDB Remote Serial Protocol (RSP) over TCP/IP.

• Why gdb-multiarch is Required: The standard host 'gdb' binary is compiled strictly to decode x86_64 ISA instructions, register sets (rax, rip, rsp), and ABI calling conventions. Because the target binary is 32-bit ARMv7-A EABI5 (using r0-r15, cpsr, Thumb/ARM instructions), gdb-multiarch is required to parse ARM ELF headers, disassemble ARM machine code, and interpret ARM register layouts.

Question 2 (Valgrind Leak Categories & Memory Error Detection):

• 'Definitely Lost' vs. 'Indirectly Lost': 'Definitely lost' means no valid pointer points to the allocated heap block anymore; the program cannot possibly free it. 'Indirectly lost' means the block is still pointed to by another heap structure (e.g., a child node or member pointer), but the parent structure itself is 'definitely lost'.

• Classification of r->name in leaky_app.c: In delete_record(Record *r), the program explicitly calls 'free(r)' without freeing 'r->name' and 'r->scores' first. Because the parent struct 'r' WAS freed (deallocated back to the heap manager) rather than leaked, the pointers pointing to 'r->name' and 'r->scores' were destroyed. Consequently, Valgrind classifies 'r->name' (2,100 bytes) and 'r->scores' (2,000 bytes) as DEFINITELY LOST (not indirectly lost, which would only happen if 'free(r)' itself were omitted).

• Other Memory Errors Detected by Valgrind Memcheck:

1. Invalid Read / Invalid Write (e.g., heap buffer overflow, array out-of-bounds, or Use-After-Free access).
2. Conditional jump or move depending on uninitialised value(s) (using stack/heap variables before initialization).
3. Invalid/Mismatched Free (e.g., Double Free, calling free() on a stack address, or mixing new[] with free()).
4. Overlapping source and destination memory blocks in memcpy()/strcpy().

Question 3 (Optimizing heavy_compute() Based on perf Profile):

• Optimization Method 1 — Loop Invariant Hoisting / Precomputation & Memoization:

- Mechanism: In main(), heavy_compute(darr, 100000) is called 50 times in a row on the exact same unmodified array 'darr'. Computing sqrt(darr[i]) * log(darr[i] + 1.0) 5,000,000 times is redundant. By caching the computed values in a lookup array during the first pass (or hoisting the calculation outside the 50-iteration loop), subsequent iterations only perform simple additions.
- Estimated Improvement: ~98% reduction in FPU workload (nearly 50x speedup for the compute phase).

• Optimization Method 2 — SIMD Vectorization (ARM NEON / x86 AVX2) & Loop Unrolling:

- Mechanism: Instead of scalar processing one double/float element per instruction (SISD), using ARMv7 NEON SIMD registers (128-bit Q0-Q15 registers) combined with 4x loop unrolling and compiler flags (-O3

-ffast-math -ftree-vectorize) processes multiple elements in parallel per clock cycle, reduces loop branch overhead, and pipelines sequential L1 data cache prefetches.
- Estimated Improvement: 2.5x to 4x speedup on single-precision/vectorized math routines.

Question 4 (Causes of Latency Spikes in Standard Kernel & PREEMPT_RT Solutions):
• Cause 1: Non-Preemptible Hardware Interrupt Handlers (Top-Halves / Hard IRQs).
- Standard Kernel: When disk I/O or network packets arrive, the CPU disables interrupts and runs the ISR immediately, preempting even SCHED_FIFO priority 80 user threads.
- PREEMPT_RT Fix: Force-threads almost all interrupt handlers into kernel threads (irq/XX-name) that can be prioritized below critical real-time user tasks.
• Cause 2: Non-Preemptible Kernel Critical Sections (Raw Spinlocks & RCU).
- Standard Kernel: Whenever a kernel thread acquires a normal spinlock_t, kernel preemption is disabled across that CPU core until the lock is released.
- PREEMPT_RT Fix: Converts spinlock_t into preemptible sleeping locks (rt_mutex) with Priority Inheritance, allowing high-priority tasks to preempt low-priority lock holders.
• Cause 3: Memory Management Overhead (Page Faults, TLB Shootdowns, & Cache Eviction).
- Standard Kernel: Dynamic page allocation or swapped-out pages trigger major/minor page faults inside the kernel MMU handler.
- PREEMPT_RT + Application Fix: Combining '--mlockall' (mlockall(MCL_CURRENT | MCL_FUTURE)) with PREEMPT_RT deterministic allocator paths prevents page faults during real-time execution.

Question 5 (Selecting the Optimal Debugging Tool for Real-World Scenarios):

Scenario	Selected Tool	Technical Justification
(a) Random application crash after several hours of uptime	GDB / gdbserver (with Core Dump analysis)	Random crashes (SIGSEGV/SIGABRT) require inspecting the exact instruction pointer (pc), register state, and call stack (backtrace) at the moment of failure via gdbserver or post-mortem core dumps.
(b) Application becomes progressively slower over time	Valgrind (Memcheck / Massif)	Progressive degradation over hours is a classic symptom of a heap memory leak exhausting physical RAM (VmRSS) and triggering continuous swap/paging thrashing. Valgrind pinpoints the exact unfreed malloc() call sites.
(c) Control system occasionally misses critical deadlines	cyclictest (and ftrace/rtla)	Occasional deadline misses are caused by worst-case OS scheduling/interrupt latency spikes rather than average throughput. cyclictest measures Max latency under load to verify real-time determinism.
(d) A specific function slows down the entire system	perf (and strace if I/O-related)	perf record/report uses low-overhead hardware PMU

		sampling to generate call-graph percentages and Flame Graphs, immediately isolating the exact CPU hotspot function and assembly instructions.

7. Practical Challenges Encountered and Solutions

#	**Issue / Challenge Encountered**	**Root Cause Analysis**	**Implemented Solution**
1	gdbserver failed with 'Temporary failure in name resolution' inside QEMU	When invoking 'gdbserver :1234', gdbserver attempted to resolve the empty host/IPv6 wildcard using DNS resolver libraries missing in the minimal BusyBox initramfs.	Explicitly specified the IPv4 wildcard address: 'gdbserver 0.0.0.0:1234 /usr/bin/buggy_app', bypassing DNS lookup and binding directly to TCP port 1234.
2	Hardware Watchpoint error in QEMU ('Target does not support this type of hardware watchpoint')	QEMU vexpress-a9 software emulation does not emulate ARMv7 hardware debug watchpoint registers for ptrace inside the guest.	Switched GDB to software watchpoints via 'set can-use-hw-watchpoints 0' before executing 'watch i', allowing GDB to single-step and catch 'i == 5' accurately.
3	cyclictest failed with 'opening histogram file:: No such file or directory' under sudo	Passing '--histfile=~/embedded_lab3/logs/hist_standard.txt' under sudo expanded '~' to '/root' instead of '/home/phuc', where '/root/embedded_lab3/logs' did not exist.	Replaced the tilde path with the explicit absolute path '/home/phuc/embedded_lab3/logs/hist_standard.txt' and refined the awk script with '!/^#/' to ignore comment footers.